MAPLE & MAIN · CUSTOMER RETENTION

Save the right customers. Stop paying the wrong ones.

A churn model, a measured view of who responds to retention offers, and a plan that lifts CRM profit by 104% on the same budget.

Synthetic data. No client, employer or proprietary information.

EXECUTIVE SUMMARY

The churn model works. Using it to pick who gets an offer does not.

$5.2M

of annual margin sits with the 17,600 active customers we expect to lapse in the next six months.

3.6x

The model's top 10% churn at 3.6x the average rate. But the highest-risk customers respond to offers at about half the rate of mid-risk customers.

$251k

more profit a year from the same 325,000 contacts, by ranking on value at risk and uplift instead of sending at random.

Recommendation: keep the churn model for sizing and early warning, pick offer audiences by measured uplift and value, and stop sending discounts to the most loyal 10% of customers, who cost $99k a year in offers they did not need.

APPROACH

Four questions, answered in order

Most churn projects stop at the second box. The money is in the third and fourth.

1. What counts as churn?

Maple & Main has no contracts, so churn has to be inferred from silence. We tested five inactivity windows.

SQL · 2.3M orders

2. Who is at risk?

A model scores every active customer using only what was known on the scoring date, tested on a later period.

SQL features · Python and SAS models

3. Who can be persuaded?

Past campaigns kept a 10% holdout. That lets us measure what an offer changes, not just who buys.

Uplift model · 625,000 contacts

4. What is it worth?

Incremental margin, less the discount given to every responder, less contact cost. Compared across targeting rules.

Back-test on nine later campaigns

DATA USED	CHECKS BEFORE MODELLING	HOW RESULTS WERE VALIDATED
200,000 loyalty members, three years of orders, web activity, service contacts, 38 CRM campaigns	Seven data-quality issues found and fixed, including 46,521 mixed-format dates and 2,329 duplicate orders	Out-of-time test, randomised holdouts, and the generator's hidden ground truth

DEFINING CHURN

180 days of silence is the working definition, and it is wrong 42% of the time

NO ORDER FOR	FLAGGED AS CHURNED	FLAGGED WHO TRULY LEFT	TRUE LEAVERS CAUGHT
60 days	36%	22%	89%
90 days	27%	31%	82%
120 days	21%	41%	75%
180 days	**16%**	**59%**	**65%**
240 days	13%	74%	56%

Active customers on 3 November 2025. "Truly left" uses the hidden churn date in the synthetic data, which a real project would not have.

A short window flags quickly but mostly catches customers who were only between purchases. A long window is accurate but arrives too late to act.

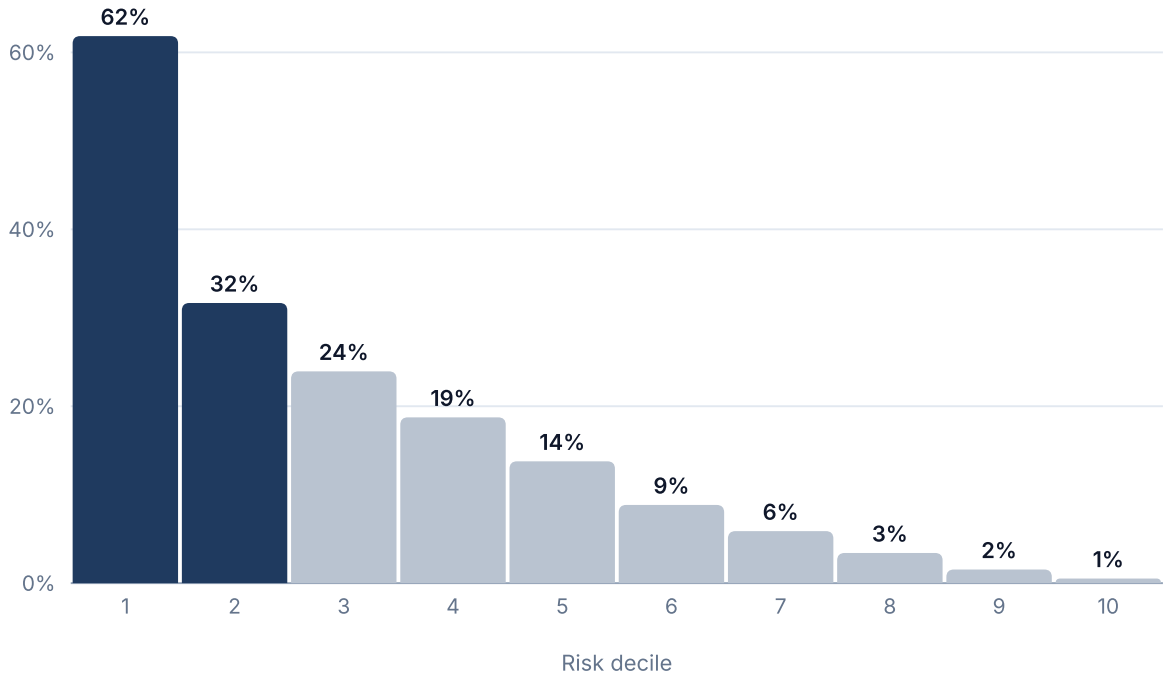
180 days is the compromise: about six in ten flagged customers have really gone, and two thirds of real leavers are caught.

> Infrequent buyers drive the false alarms. Treat a "churned" flag as a probability, not a fact, and never write customers off on the flag alone.

MODEL PERFORMANCE

The top 20% of scores contain 55% of the customers who go quiet

Churn rate by risk decile, May 2026 test snapshot



Decile 1 is the highest predicted risk. Average churn rate is 17.0% across 112,999 active customers.

APPROACH	AUC	TOP-DECILE LIFT
Recency rule (longest since last order)	0.69	2.3x
Logistic scorecard (SQL + SAS)	0.81	3.3x
Gradient boosting, calibrated	**0.83**	**3.6x**

- Trained on four snapshots from 2024 and 2025, tested once on a snapshot nine months later.
- Predicted and actual churn rates agree within four points in every score band.
- Score distribution is stable between training and test (PSI 0.004).
- Against the hidden truth the model scores 0.78: it finds real leavers, not just quiet customers.

DRIVERS AND VALUE

Fading web activity is the earliest warning, and risk is concentrated in a small group

What the model relies on most



Loss in AUC when each input is shuffled. Loyalty tier is a current-state field here; a production build would use tier history.

$5.2M

12-month margin at risk, out of $59.9M from active customers.

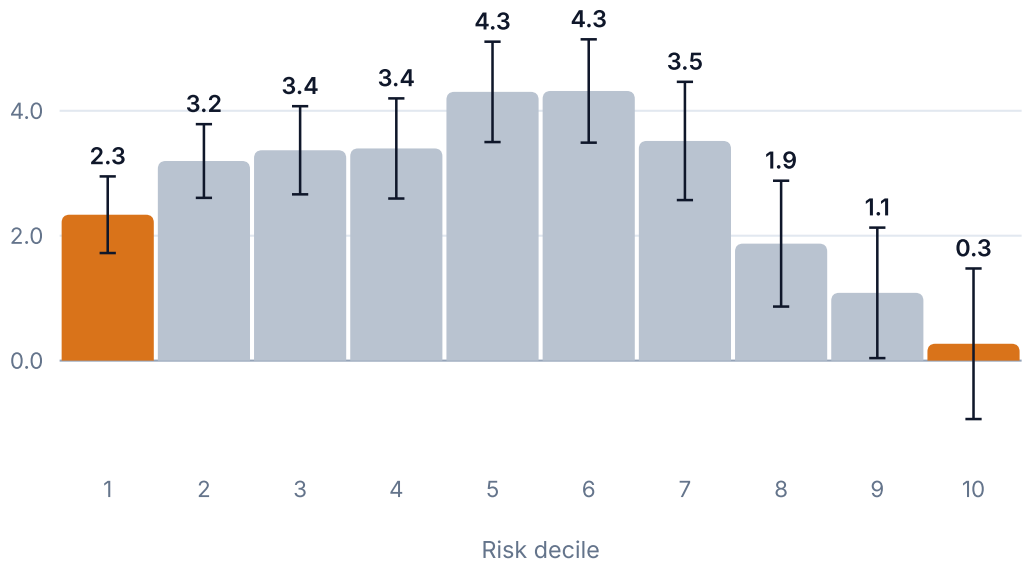
47%

of that margin at risk sits with just 10% of customers. Risk alone does not find them: value has to be in the ranking.

THE FINDING THAT CHANGES THE PLAN

Offers work best on mid-risk customers and lose money on the most loyal

Extra customers who order because of the offer, by churn risk



Percentage points, treated minus holdout, nine campaigns from February to October 2026. Bars show 95% intervals. Decile 1 is the highest risk.

Net profit per contact, by churn risk



Incremental margin less offer cost less contact cost, CAD. Offer cost is an assumption: 4% to 20% of the basket depending on the offer.

The highest-risk customers have mostly decided already: an offer moves 2.3 points of them, against 3.5 in the middle. The lowest-risk customers were going to buy anyway, so the discount is simply given away: -$3.05 per contact.

A LIMIT WORTH KNOWING

Offers buy an order. They do not buy loyalty.

87.4%

of customers who received an offer were still buying in the following six months.

87.3%

of the holdout customers, who received nothing, were too.

Across 19 campaigns with a full follow-up window, the difference is 0.05 points, with a margin of error of ±0.32. That is no measurable effect.

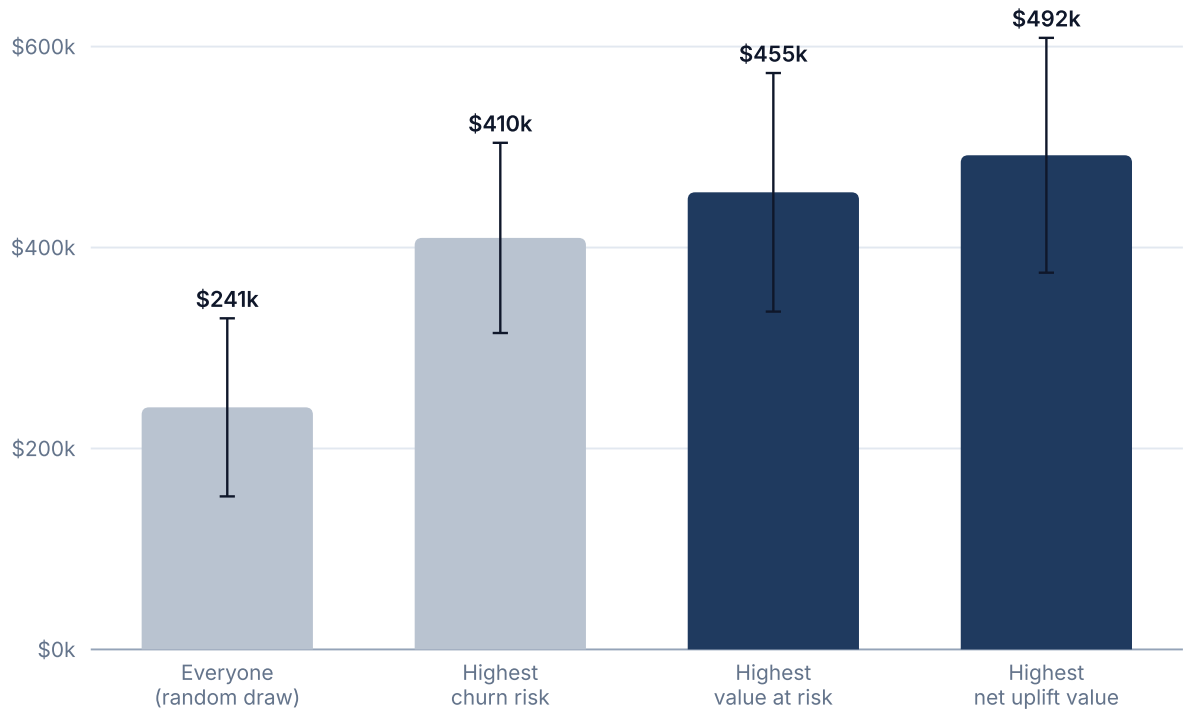
So this business case counts only the margin from the extra order in the 14-day response window. It assumes nothing about customers staying longer.

> If retention is the goal, discounts are the wrong tool. The service data points somewhere else: complaints and unresolved issues are among the model's top warning signs, and they are fixable.

BUSINESS CASE

Same contacts, better list: $251k more profit a year

Annual net profit from CRM offers, by how the audience is chosen



CAD. 25,000 contacts per campaign, 13 campaigns a year. Measured on nine held-out campaigns and scaled to a year. Bars show 95% intervals.

- **A random draw is the worst option.** Every rule that uses what the data already knows beats it.
- **Risk multiplied by customer value captures most of the gain** and needs no extra model. Start there.
- **The uplift model edges ahead** and adds one thing the simple ranking cannot: a do-not-contact list. The two are within each other's margin of error.

> Honest scale: the gain is 0.4% of annual margin. It is real and costs nothing to capture, but offers alone will not move retention.

Contacting more of the base would also add profit, since the average contact pays back. That depends on customers tolerating more frequent offers, which this data cannot show, so any volume increase should be tested with a holdout.

RECOMMENDATIONS

Three actions, in order of effort

1 **Stop sending offers to the most loyal 10%** — **$99k a year**

They buy at the same rate with or without a discount. A suppression rule in the campaign tool is enough.

2 **Choose audiences by value at risk, then by uplift** — **$251k a year**

Score the whole eligible base each month and take the top 25,000, instead of a random draw. Keep the 10% holdout so the result stays measurable.

3 **Test a service-recovery play for high-risk customers** — **To be measured**

Offers do not hold them. Fast resolution of open complaints might. Run it as an experiment with its own holdout before scaling.

The first two overlap: the better list already leaves out most of the loyal 10%, so the two figures should not be added together.

APPENDIX

Method, assumptions and limits

Population. Customers who joined at least 30 days before the snapshot and ordered in the prior 180 days.

Features. 33 inputs built in SQL from data dated before the snapshot: purchase recency, frequency and value, web activity, service contacts, CRM history, profile.

Validation. Train on four snapshots (Nov 2024 to Aug 2025), calibrate on Nov 2025, test on May 2026.

Uplift. Single gradient-boosting model with the treatment flag as an input, chosen over a two-model design on earlier campaigns. Predicted uplift tracks the true uplift built into the data (correlation 0.58).

Back-test. Each targeting rule is scored only on measured treated-versus-holdout differences in nine campaigns the models never saw.

Assumed, not measured. Offer cost as a share of the responder's basket: 10% off 10%, bonus points 4%, $15 off $75 12%, free shipping 5%, win-back 20% off 20%. The simulator lets you change it.

Limits. Contacting the same top customers every month may tire them; this is not modelled. Loyalty tier is not point-in-time. Intervals reflect holdout sampling only.

Synthetic data. Maple & Main is fictional. The data was generated with known churn, uplift and media mechanisms, which is why results can be checked against ground truth. Figures illustrate the method, not any real company.

Code. SQL, Python and SAS, with a model card and QC log, are in the project repository.